Cart lab: change mass

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a controlled comparison to describe the effect of system mass on acceleration.

SUCCESS CRITERIA - I can...

1. Identify total moving mass and keep driving force fixed.
2. Compare acceleration values using a fair test.
3. Separate model data from measured evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “mass” and “system” to explain your model or result. Sentence starters are optional.

mass: a measure of how strongly an object resists acceleration

system: the object or objects chosen for analysis

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

A cart accelerates uniformly from rest to 12 m/s east in 4 s.
Find acceleration, average velocity and displacement.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

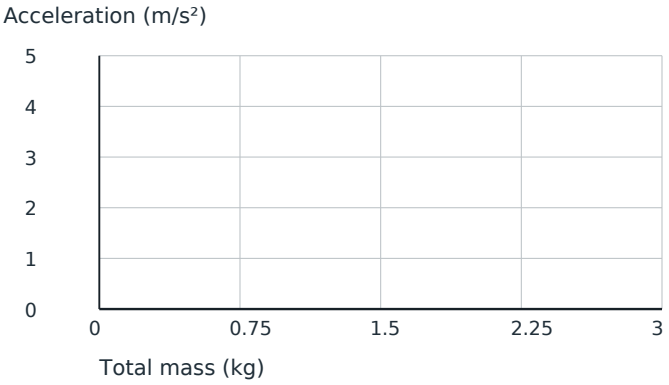
02 Worked example: explain the model

Reference: $a = F_{net}/M = m_h\, g/M$.
Ideal cart + hanger: M is all moving mass; m_h is hanging mass. Hold m_h fixed; neglect resistance and pulley inertia.

Worked reasoning: Supplied cart trials use the same net force: masses 1 and 2 kg accelerate at 4 and 2 m/s². In both cases ma=4 N; doubling mass halves acceleration.

Your turn: Plot acceleration against mass for the two trials.
Label what was held constant.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read the sample pattern

The table gives ideal cart trials with the same 4 N net force.
Compare the 1 kg and 2 kg trials. What happens to acceleration when mass doubles? Support your claim with both values.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .

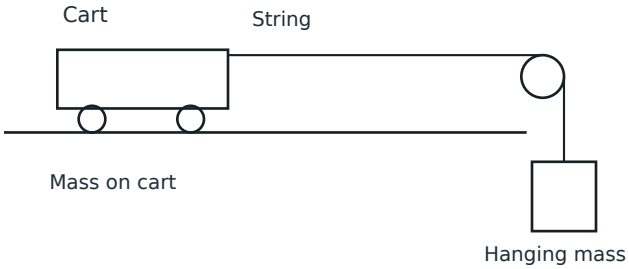
M (kg)	a (m/s²)
1	4
2	2
4	1

Marzano check - circle: ① ② ③

04 Plan a fair test

In the cart-and-hanger setup, you will change total moving mass while keeping the driving force the same.
Label what you change and what you measure.
Name two controls. Why would adding mass to the hanger spoil this comparison?

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: ① ② ③

05 Record evidence honestly

If equipment is available, follow the teacher's setup and record three acceleration trials at each of two total masses. Otherwise use the supplied model data in 3 and label it model data.
Compare the results.

Support: My claim is ____. My two supporting values are ____.

Sentence starter: Both … . They differ in … because … .

Total M	Trial 1	Trial 2	Trial 3
____	____	____	____
____	____	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A 4 kg sign hangs still from one vertical cord.
Use g=10 N/kg.
Name both forces, their sources, magnitudes and directions.
Draw the forces on the chosen object and label their sources.

Sentence starter: The … in my model represents … . Its label shows … .



Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

At the same net force, a 1 kg system accelerates at 6 m/s².
Predict acceleration for a 2 kg system.
Name the condition that makes the prediction valid.

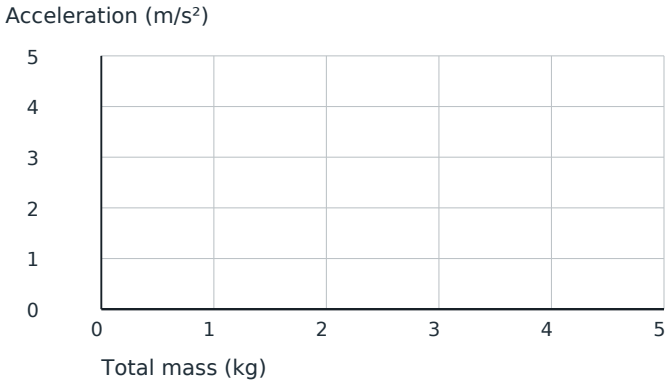
Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2.
Predict acceleration for a 4 kg cart at the same net force. Then describe a fair trial that could challenge your prediction.

Sentence starter: I predict … . My model supports this because … .



Marzano check - circle: 1 2 3